

Dummy-Variable Regression

for categorical predictors

Ms. Signal 🧑🏫 Mr. Noise





Recap

Last week

Multiple regression

- ❑ Partial slope coefficient (“having controlled for”)
- ❑ Adjusted R^2
- ❑ Standardization (for comparing slopes)

Moderation / interaction analysis

- ❑ Simple slopes analysis (slope of X_1 depends on X_2)
- ❑ Centering (for meaningful slope coefficients)

Last year

Four levels of measurement

- ❑ Nominal scale: 🍏 🍓 🍌
- ❑ Ordinal scale: 😞 😐 😊
- ❑ Interval scale: 📅 July 17 🌡️ (°C)
- ❑ Ratio scale: 🕒 📏 ⚖️ 🌡️ (K)

Two types of variables

- ❑ Categorical
 - ❑ Dichotomous: ♂ ♀
 - ❑ Polytomous: 🍏 🍓 🍌 🍎
- ❑ Numerical
 - ❑ Discrete: 🎲
 - ❑ Continuous: 🌡️



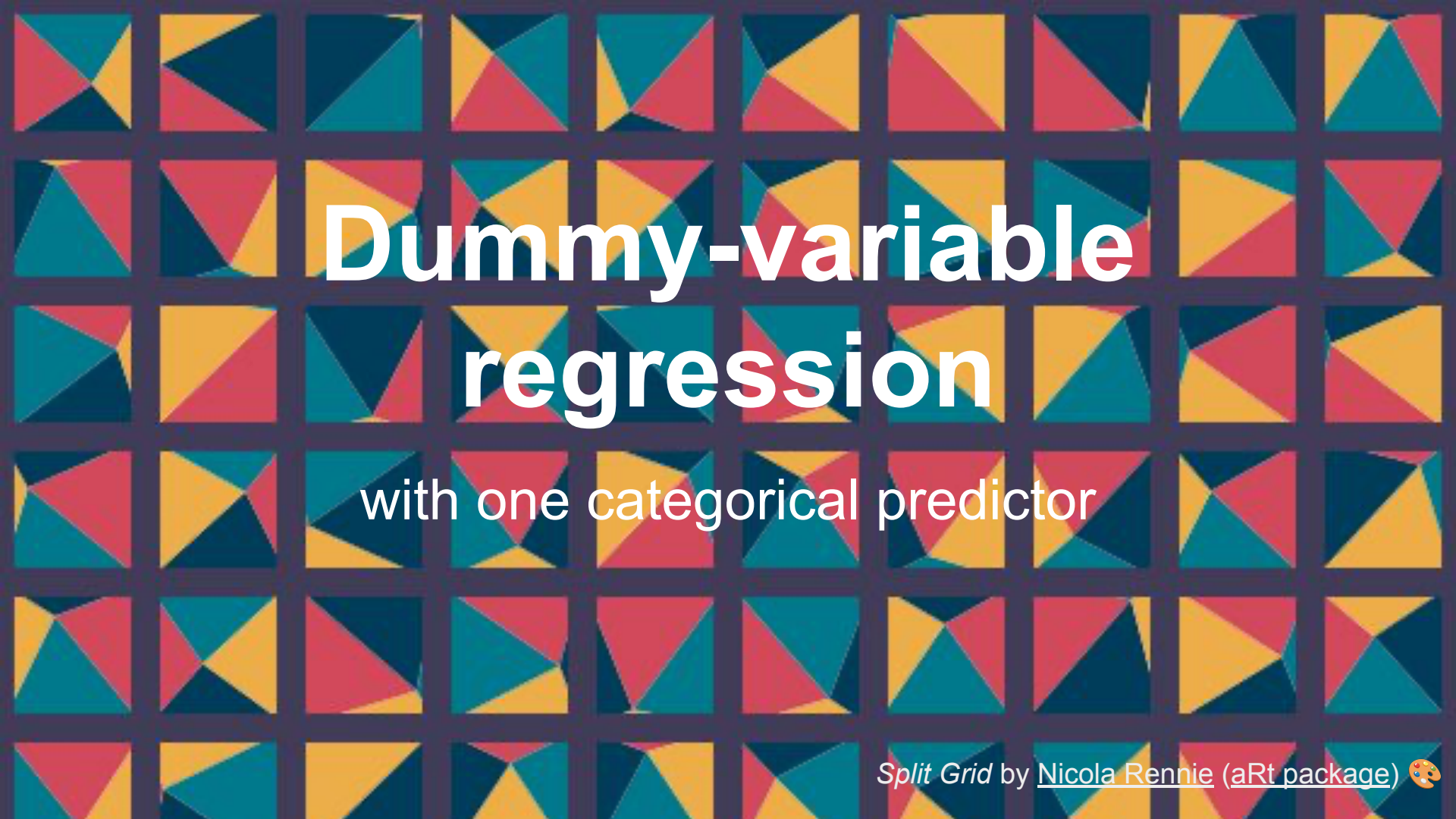
Today

Topics

- 1 | Statistical reasoning with GLM
- 2 | Multiple linear regression
- 3 | Dummy-variable regression
 - 3.1 | One predictor
 - 3.2 | Two predictors
 - 3.3 | One predictor, one moderator
- 4 | Bayesian statistics
- 5 | Logistic regression
- 6 | Multilevel and longitudinal analysis
- 7 | Statistics superpowers

Learning goals

- 😄 Test the difference between several categories (e.g., experimental conditions).
- 😄 Use categorical independent variables in a prediction model.
- 😄 Control for categorical independent variables.



Dummy-variable regression

with one categorical predictor

Student evaluations | Sex

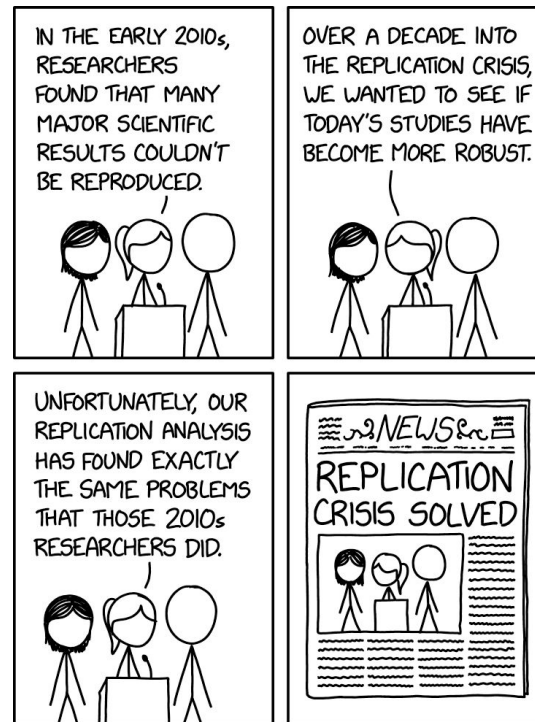
De student als consument maakt vrouwelijke docenten extra kwetsbaar

Nieuws | door Frans van Heest

13 september 2023 | Vrouwelijke docenten worden aantoonbaar gediscrimineerd door studentenevaluaties, maar toch blijft het instrument voor veel universiteiten belangrijk om medewerkers te beoordelen. Cursusevaluaties moedigen echter middelmatig onderwijs aan en zijn extra nadelig voor vrouwen.

sex 📊

rating 📈



Categorical independent variables



*I gave each category a number (male = 0, female = 1),
and look, no errors!*

```
mod <- score ~ gender
dat <- dat |> mutate(gender = ifelse(gender ==
"male", 0, 1))
summary(lm(mod, data = evals))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.23433	0.03298	128.384	< 2e-16 ***
gender	-0.14151	0.05082	-2.784	0.00558 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5399 on 461 degrees of freedom
Multiple R-squared: 0.01654, Adjusted R-squared: 0.01441
F-statistic: 7.753 on 1 and 461 DF, p-value: 0.005583

I guess you're right, but will it always? 

```
t.test(mod, data = evals, var.equal = TRUE)
```

Two Sample t-test

```
data: score by gender
t = -2.7844, df = 461, p-value = 0.005583
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -0.2413779 -0.0416378
sample estimates:
mean in group female mean in group male
 4.092821           4.234328
```



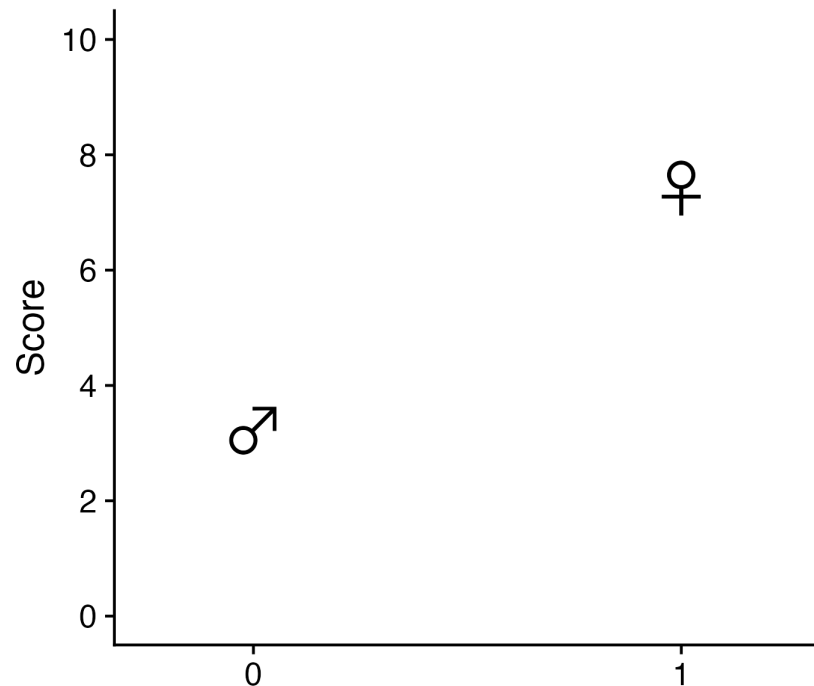
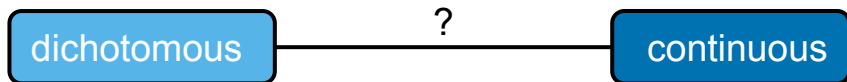
Dummies for dummies | Dichotomous

Dichotomous variable

♂ = 0

♀ = 1

Why does it work?





Dummies for dummies | Polytomous ❌

Polytomous variable

🍏 = 0

🍓 = 1

🍌 = 2

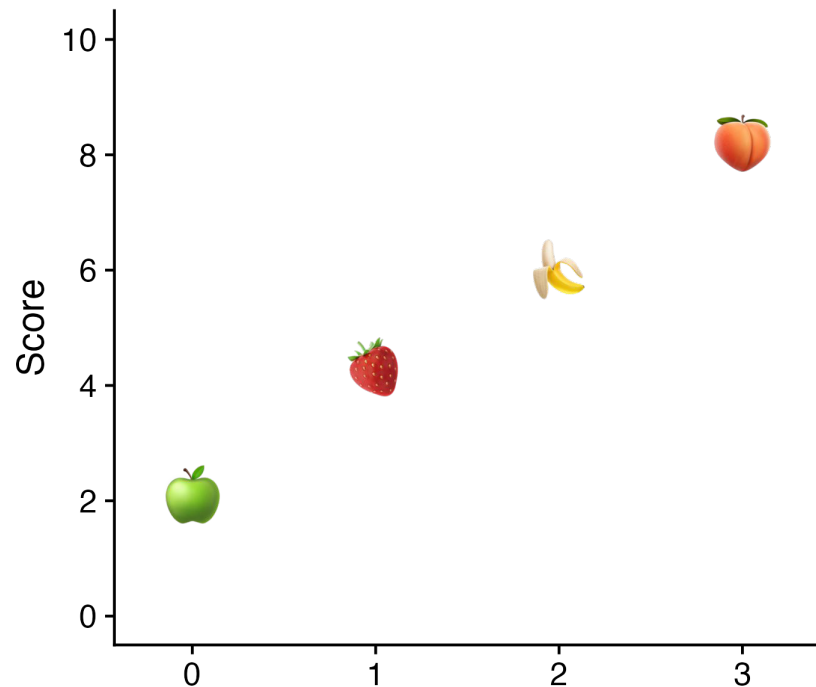
🍑 = 3

Why does it not work?

polytomous

?

continuous





Dummies for dummies | Dummy-coding

Dummy-coding

Original data

Y	X
5	
6.2	
2	
4.7	

Dummy-coded data

Y				
5	1	0	0	0
6.2	0	1	0	0
2	0	0	1	0
4.7	0	0	0	1

So, you basically dummy-coded sex 

 *That's called statistical intuition, you should use that some time*

No need to be rude, Mr. Mansplain 



Factor

```
class(data$categorical)
data$categorical <- factor(data$categorical)
```

Student evaluations | Academic rank



Teacher



Tenure track



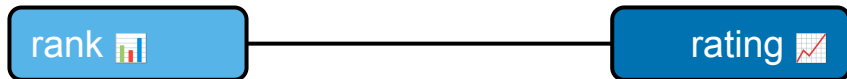
Tenured

rank 

rating 

Model thinking

Do professors with different academic ranks receive different ratings?



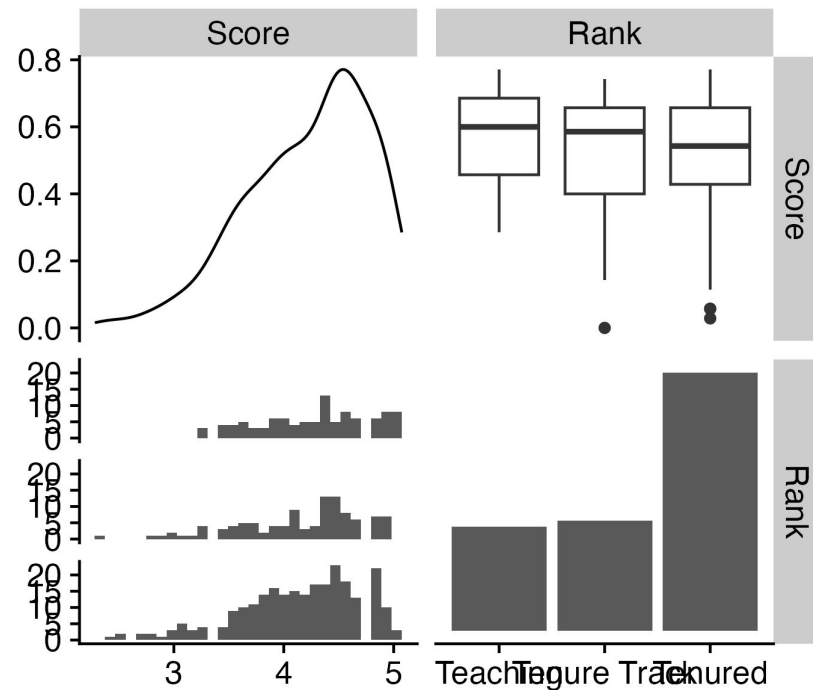
$$\text{score} = \beta_0 + \beta_1(\text{rank}_{\text{tenure track}}) + \beta_2(\text{rank}_{\text{tenured}}) + \epsilon$$

```
mod <- score ~ rank
```

		Independent	
		Categorical	Continuous
Dependent	Cate gori cal		
	Con tinu ous	Dummy-variable regression	Simple regression Multiple regression

Data | Transformation & visualization

```
data(evals)
class(evals$rank)
levels(evals$rank)
summary(evals$rank)
GGally::ggpairs(evals, columns = c("score",
"rank"))
```



Results | With & without intercept

```
mod <- score ~ rank
fit <- lm(mod, data = evals)
summary(fit)
```

```
Call:
lm(formula = mod, data = evals)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8546 -0.3391  0.1157  0.4305  0.8609
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.28431    0.05365   79.853  <2e-16 ***
ranktenure track -0.12968    0.07482  -1.733   0.0837 .
ranktenured    -0.14518    0.06355  -2.284   0.0228 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5419 on 460 degrees of freedom
Multiple R-squared:  0.01163, Adjusted R-squared:  0.007332
F-statistic: 2.706 on 2 and 460 DF, p-value: 0.06786
```

$$\widehat{\text{score}} = 4.28 - 0.13(\text{rank}_{\text{tenure track}}) - 0.15(\text{rank}_{\text{tenured}})$$

```
mod <- score ~ 0 + rank
fit <- lm(mod, data = evals)
summary(fit)
```

```
Call:
lm(formula = mod, data = evals)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8546 -0.3391  0.1157  0.4305  0.8609
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
rankteaching    4.28431    0.05365   79.85  <2e-16 ***
ranktenure track  4.15463    0.05214   79.68  <2e-16 ***
ranktenured      4.13913    0.03407  121.50  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5419 on 460 degrees of freedom
Multiple R-squared:  0.9835, Adjusted R-squared:  0.9834
F-statistic: 9163 on 3 and 460 DF, p-value: < 2.2e-16
```

$$\widehat{\text{score}} = 4.28(\text{rank}_{\text{teaching}}) + 4.15(\text{rank}_{\text{tenure track}}) + 4.14(\text{rank}_{\text{tenured}})$$



For dummies dummies | Changing the reference group

```
evals$rank <- relevel(evals$rank, ref = "tenure
track")
levels(evals$rank)
```

```
Call:
lm(formula = mod, data = evals)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8546 -0.3391  0.1157  0.4305  0.8609
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.15463    0.05214   79.680  <2e-16 ***
rankteaching   0.12968    0.07482    1.733   0.0837 .
ranktenured   -0.01550    0.06228   -0.249   0.8036
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5419 on 460 degrees of freedom
Multiple R-squared:  0.01163, Adjusted R-squared:  0.007332
F-statistic: 2.706 on 2 and 460 DF, p-value: 0.06786
```



Linear combinations (see next slide)

- ☐ $\beta_{\text{Teaching}} - \beta_{\text{Tenure Track}}$
- ☐ $\beta_{\text{Tenured}} - \beta_{\text{Tenure Track}}$
- ☐ $\beta_{\text{Tenured}} - \beta_{\text{Teaching}}$



Family-wise error rate = $1 - (1 - \alpha)^m$

- ☐ $m = 3$
- ☐ $\alpha = .05$
- ☐ $\text{FWER} \approx .14$

Bonferroni adjustment = $\alpha / m \approx .017$, but power!

Results | Pairwise multiple comparison adjustment

```
tukey <- glht(fit, linfct = mcp(rank = "Tukey"),  
vcov = sandwich)  
summary(tukey)
```

Simultaneous Tests for General Linear Hypotheses

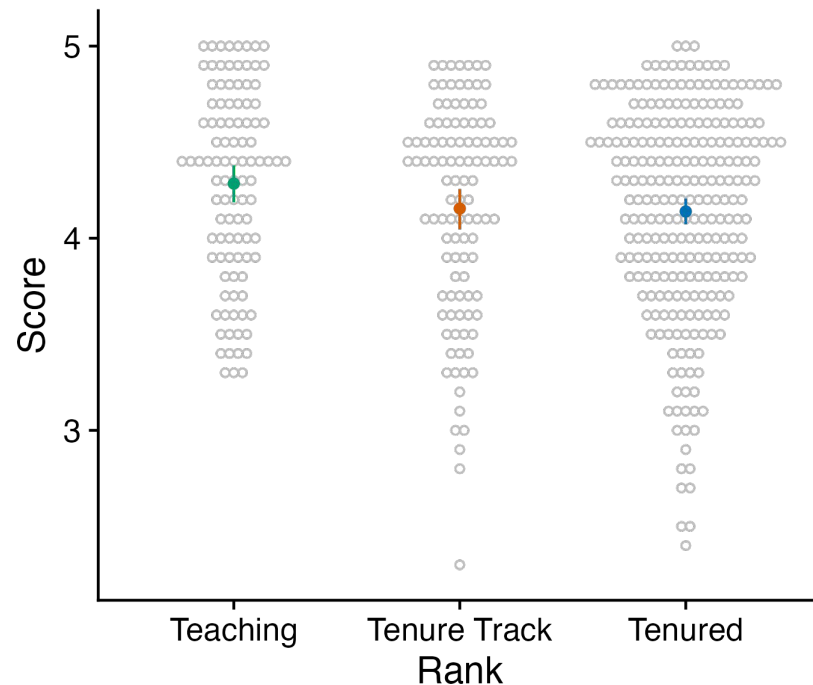
Multiple Comparisons of Means: Tukey Contrasts

Fit: lm(formula = mod, data = evals)

Linear Hypotheses:

	Estimate	Std. Error	t value	Pr(> t)
teaching - tenure track == 0	0.12968	0.07279	1.782	0.1753
tenured - tenure track == 0	-0.01550	0.06388	-0.243	0.9678
tenured - teaching == 0	-0.14518	0.06002	-2.419	0.0417 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)



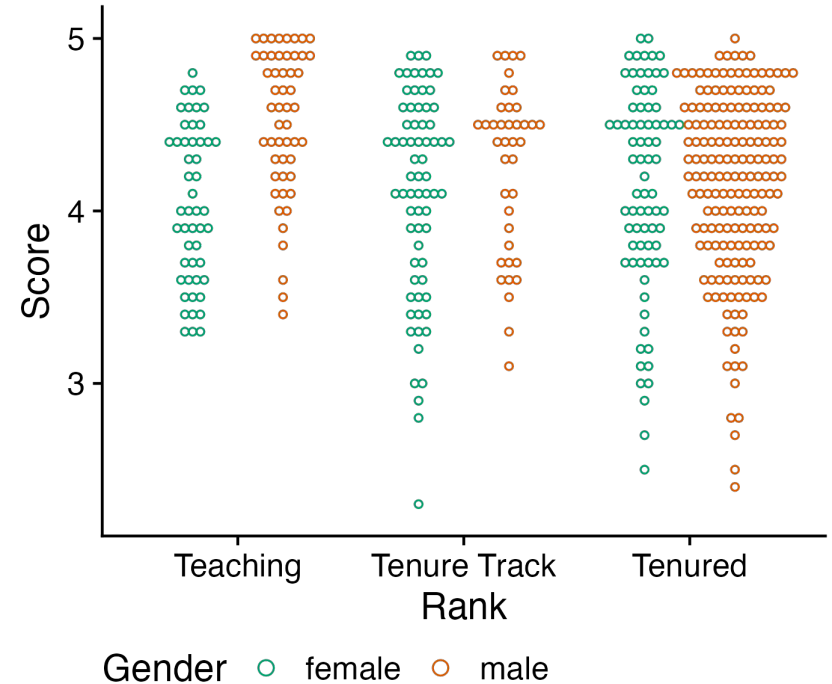


Dummy-Variable Regression

with more than one categorical predictor

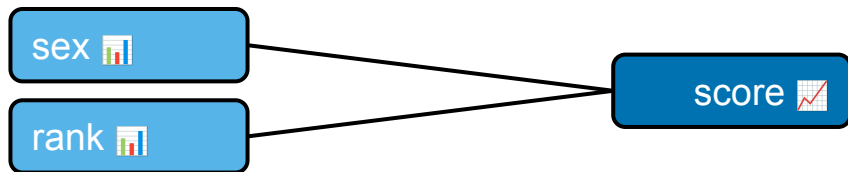
Student evaluations | Gender + rank

- Male > Female
- Teaching > Tenured



Model thinking

Do professors with different academic ranks receive different ratings, given their sex?



$$\text{score} = \beta_0 + \beta_1(\text{gender}_{\text{male}}) + \beta_2(\text{rank}_{\text{tenure track}}) + \beta_3(\text{rank}_{\text{tenured}}) + \epsilon$$

```
mod <- score ~ gender + rank
```

To which group refers score, when...

- ☐ $\beta_1 = \beta_2 = \beta_3 = 0$
- ☐ $\beta_1 = 1$
- ☐ $\beta_1 = \beta_2 = 1$
- ☐ $\beta_3 = 1$

Results

```
fit <- lm(mod, data = evals)
summary(fit)
```

```
Call:
lm(formula = mod, data = evals)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.7941 -0.3418  0.1011  0.4105  0.9781
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.19887    0.05954   70.520 < 2e-16 ***
gendermale      0.16760    0.05272    3.179  0.00158 **
ranktenure track -0.10476    0.07450   -1.406  0.16033
ranktenured     -0.17699    0.06373   -2.777  0.00570 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5366 on 459 degrees of freedom
Multiple R-squared:  0.03292,    Adjusted R-squared:  0.0266
F-statistic: 5.208 on 3 and 459 DF,  p-value: 0.001519
```

$$\widehat{\text{score}} = 4.2 + 0.17(\text{gender}_{\text{male}}) - 0.1(\text{rank}_{\text{tenure track}}) - 0.18(\text{rank}_{\text{tenured}})$$

- ❑ What does the *intercept* estimate represent?
- ❑ What does the *ranktenured* estimate represent?

The mean score of females in a tenure track is .10 points lower than the mean score of female teachers, having controlled for males and tenured females.

- ❑ Is there an effect *rank*, having controlled for *gender*?

Results | *F*-test

```
car::linearHypothesis(  
  model = fit,  
  hypothesis.matrix = c("ranktenure track = 0",  
                        "ranktenured = 0"))
```

Linear hypothesis test:

ranktenure track = 0
ranktenured = 0

Model 1: restricted model

Model 2: score ~ gender + rank

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	461	134.39				
2	459	132.16	2	2.2382	3.8869	0.02119 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Rank has a significant relation with *score*,
having controlled for *gender* ($\alpha = .5$).

```
car::linearHypothesis(  
  model = fit,  
  hypothesis.matrix = c("gendermale = 0"))
```

Linear hypothesis test:

gendermale = 0

Model 1: restricted model

Model 2: score ~ gender + rank

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	460	135.06				
2	459	132.16	1	2.9093	10.104	0.001579 **

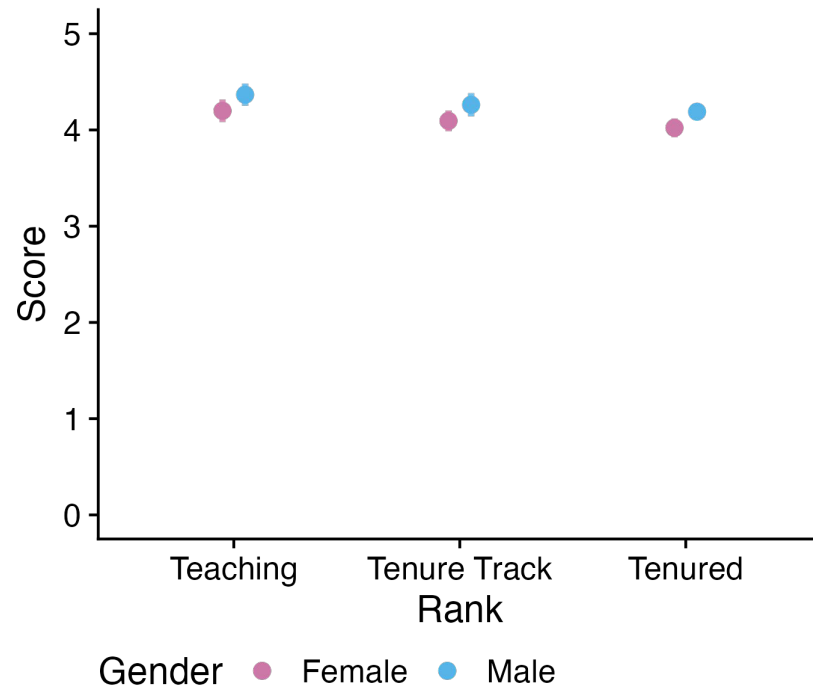
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Gender has a significant relation with *score*,
having controlled for *rank* ($\alpha = .5$).

Results | Visualization

```
?interactions::cat_plot  
cat_plot(model = fit,  
  pred = "rank",  
  modx = "gender",  
  geom = "point",  
  interval = TRUE) +  
  ylim(0, 5)
```

- Relationship between score and gender is restricted to be the same across ranks (and vice versa). I.e., we used an additive model.

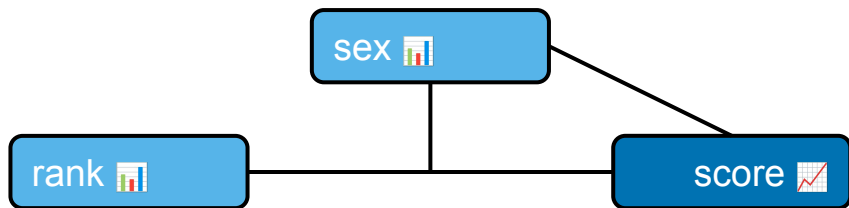


Dummy-variable regression

with a categorical predictor and a categorical moderator

Student evaluations | Gender $\times$ rank

Is the effect of academic rank on rating moderated by sex?



$$\text{score} = \beta_0 + \beta_1(\text{gender}_{\text{male}}) + \beta_2(\text{rank}_{\text{tenure track}}) + \beta_3(\text{rank}_{\text{tenured}}) + \beta_4(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenure track}}) + \beta_5(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenured}}) + \epsilon$$

```
mod <- score ~ gender * rank
```

Results

```
fit <- lm(mod, data = evals)
summary(fit)
```

```
Call:
lm(formula = mod, data = evals)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.79710	-0.34520	0.07885	0.37885	0.87500

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.03800	0.07498	53.855	< 2e-16 ***
gendermale	0.48315	0.10501	4.601	5.46e-06 ***
ranktenure track	0.05910	0.09847	0.600	0.548660
ranktenured	0.08700	0.09654	0.901	0.367982
gendermale:ranktenure track	-0.32385	0.14936	-2.168	0.030660 *
gendermale:ranktenured	-0.46296	0.12773	-3.625	0.000322 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5302 on 457 degrees of freedom
Multiple R-squared: 0.05996, Adjusted R-squared: 0.04967
F-statistic: 5.83 on 5 and 457 DF, p-value: 3.11e-05

$$\widehat{\text{score}} = 4.04 + 0.48(\text{gender}_{\text{male}}) + 0.06(\text{rank}_{\text{tenure track}}) + 0.09(\text{rank}_{\text{tenured}}) \\ 0.32(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenure track}}) - 0.46(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenured}})$$

```
sim_slopes(model = fit, pred = "gender",
            modx = "rank", johnson_neyman = FALSE)
```

SIMPLE SLOPES ANALYSIS

Slope of gender when rank = teaching:

Est.	S.E.	t val.	p
0.48	0.11	4.60	0.00

Slope of gender when rank = tenure track:

Est.	S.E.	t val.	p
0.16	0.11	1.50	0.13

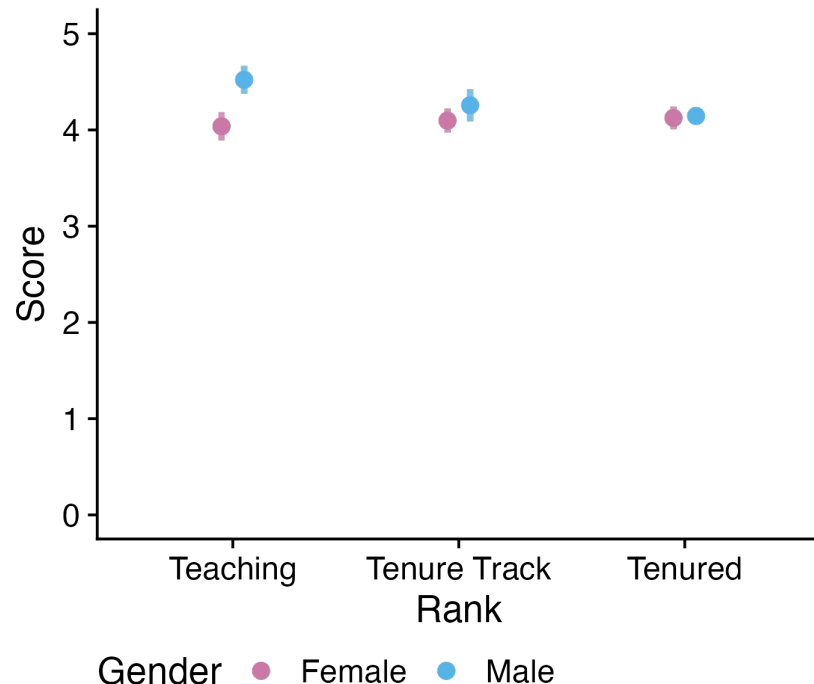
Slope of gender when rank = tenured:

Est.	S.E.	t val.	p
0.02	0.07	0.28	0.78

Results | Visualization

```
?interactions::cat_plot  
cat_plot(model = fit,  
  pred = "rank",  
  modx = "gender",  
  geom = "point",  
  interval = TRUE) +  
  ylim(0, 5)
```

- ❑ Relationship between score and gender may vary across ranks (and vice versa). I.e., we used a nonadditive model.
- ❑ Slope is no longer significant for each rank.





R is the shit

```
library("equatiomatic")
extract_eq(fit, intercept = "beta", wrap = TRUE,
use_coefs = TRUE)
```

```
$$
\begin{aligned}
\operatorname{\widehat{score}} &= 4.2 + 0.17(\operatorname{gender}_{\operatorname{male}}) - 0.1(\operatorname{rank}_{\operatorname{tenure\ track}}) - 0.18(\operatorname{rank}_{\operatorname{tenured}}) \\
\end{aligned}
$$
```

Copy-paste and [download as image](#):

$$\widehat{\text{score}} = 4.04 + 0.48(\text{gender}_{\text{male}}) + 0.06(\text{rank}_{\text{tenure track}}) + 0.09(\text{rank}_{\text{tenured}}) \\ 0.32(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenure track}}) - 0.46(\text{gender}_{\text{male}} \times \text{rank}_{\text{tenured}})$$





Colophon

Slides

alexandersavi.nl/teaching/

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